

14. **Which of these constructors is allowed?**
 - A) abstract
 - B) static
 - C) final
 - D) None of the above
15. **Constructor overloading means:**
 - A) Using super()
 - B) Having multiple constructors with different parameters
 - C) Copying parent constructor into child
 - D) None
16. **Constructor chaining means:**
 - A) One constructor calling another using this() or super()
 - B) Multiple classes sharing the same constructor
 - C) Using private constructors
 - D) Overloading constructors in the same class
17. **Default constructor is provided when:**
 - A) At least one constructor is defined
 - B) No constructor is defined by the programmer
 - C) Class has final keyword
 - D) Class is abstract
18. **What happens if you define a parameterized constructor but no default constructor?**
 - A) Compiler still provides a default constructor
 - B) Compiler throws an error if you try to call default constructor
 - C) Both A & B
 - D) None
19. **Can a constructor call another constructor in the same class?**
 - A) No
 - B) Yes, using this()
 - C) Yes, using super()
 - D) Only if constructors are private
20. **What happens if a constructor is private?**
 - A) Object creation is not possible outside the class
 - B) Inheritance fails
 - C) Compilation failure
 - D) None
21. **this() is used for:**
 - A) Calling parent constructor
 - B) Calling another constructor in the same class
 - C) Referring to parent's method
 - D) None
22. **Why must this() or super() be the first statement in a constructor?**
 - A) To ensure proper constructor chaining
 - B) JVM rule for clarity
 - C) Optimization
 - D) None
23. **Can we use both this() and super() in the same constructor?**
 - A) Yes
 - B) No, compiler error
 - C) Yes, if in second line
 - D) Depends on JVM
24. **Can parent constructor be called without super()?**
 - A) No
 - B) Yes, compiler inserts it automatically if not written
 - C) Only if parent has abstract constructor
 - D) None
25. **If parent has only parameterized constructor, subclass must:**
 - A) Do nothing
 - B) Explicitly call super(arguments)
 - C) Define default constructor
 - D) Both B & C
26. **Can super be used in static methods?**
 - A) Yes
 - B) No, because super needs an instance
 - C) Only in abstract class
 - D) None
1. **Difference between IIB and constructor?**
 - A) Both same
 - B) IIB runs before constructor, used for common init
 - C) Constructor runs before IIB
 - D) None
2. **Can we declare multiple IIBs?**
 - A) No
 - B) Yes, executed top to bottom
 - C) Yes, random order
 - D) None
3. **Execution order for constructor, iib and sib?**
 - A) Constructor → IIB → SIB
 - B) SIB → IIB → Constructor
 - C) IIB → SIB → Constructor
 - D) None
4. **When is a static block executed?**
 - A) At class loading time
 - B) At object creation time
 - C) At JVM shutdown
 - D) None
5. **If parent and child both have static blocks?**
 - A) Parent runs first, then child
 - B) Child runs first
 - C) Random
 - D) None
6. **Which component actually converts .java → .class file?**
 - A) JVM
 - B) JRE
 - C) JDK (javac compiler)
 - D) ClassLoader
7. **Which memory area stores local variables in Java?**
 - A) Heap
 - B) Stack
 - C) Method Area
 - D) Permanent Generation (PermGen)
8. **What happens when an object is created using new keyword?**
 - A. Only memory is allocated
 - B. Constructor is not called
 - C. Memory is allocated and constructor is invoked
 - D. Only reference is created
9. **Which statement is true about constructors?**
 - A. They have a return type
 - B. They can be static
 - C. They are called explicitly
 - D. They initialize objects
10. **Why can constructors not be static?**
 - A. Static methods cannot return values
 - B. Static members belong to class, constructors create objects
 - C. Static blocks run before constructors
 - D. Static keyword is not allowed in methods
11. **Which statement about private constructors is CORRECT?**
 - A. Objects cannot be created at all
 - B. Constructors cannot be private
 - C. Object creation is restricted outside the class
 - D. Class cannot be instantiated even inside
12. **Which is the main purpose of constructor chaining?**
 - A. Improve performance
 - B. Reduce memory usage
 - C. Reuse constructor initialization logic
 - D. Enable method overriding
13. **Which scenario commonly uses a private constructor?**
 - A. Inheritance
 - B. Method overloading
 - C. Singleton design pattern
 - D. Polymorphism

37. Which statement about command-line arguments in Java is correct?
 - A) They are passed as String[] args to the main method
 - B) They are always converted to integers by JVM
 - C) JVM ignores them by default
 - D) Command-line arguments must be declared as int[] args
38. What is the default value of a boolean instance variable in Java?
 - A) true
 - B) false
 - C) 0
 - D) null
39. Which of the following is true about Garbage Collection?
 - A) GC is **mandatory** and runs at fixed intervals
 - B) GC automatically removes objects with **no references**
 - C) GC can delete objects that are still in use
 - D) Developers must manually call delete to free memory
40. What happens when you call System.gc() in Java?
 - A) JVM immediately runs Garbage Collection and frees all unreferenced objects
 - B) JVM requests Garbage Collection, but execution is not guaranteed
 - C) JVM disables Garbage Collection
 - D) JVM throws a compile-time error
41. What is the default value of a local variable?
 - A) 0
 - B) false
 - C) Depends on type
 - D) No default value, must be initialized
42. Which of these is not allowed in a Java constructor?
 - A) Having the same name as the class
 - B) Returning void
 - C) Calling another constructor using this()
 - D) Access modifiers like private, public
43. To run a Java program in production, what is strictly required?
 - A) JDK only
 - B) JRE only
 - C) JVM only
 - D) JDK + JRE
44. What are JVM arguments used for?
 - A) To compile Java source code into bytecode
 - B) To pass runtime options to the JVM, such as memory settings, system properties, or garbage collection behavior
 - C) To write Java programs
 - D) To execute SQL queries
45. Can a Java developer set the heap size for a Java application?
 - A) No, JVM decides heap size automatically and it cannot be changed
 - B) Yes, using **JVM arguments** like -Xms (initial) and -Xmx (maximum)
 - C) Only by changing the Java source code
 - D) Only in JDK 17+
46. In Eclipse, which key is used to step into(going inside) a method during debugging?
 - A) F6
 - B) F5
 - C) F7
 - D) F8
47. In Eclipse, which key is used to jump to the next breakpoint during debugging?
 - A) F5 – Step Into
 - B) F6 – Step Over
 - C) F7 – Step Return
 - D) F8 – Resume / Continue to next breakpoint
48. How can you convert a String to an int in Java?
 - A) int x = Integer.parseInt("123");
 - B) int x = Integer.valueOf("123");
 - C) int x = (int)"123";
 - D) Both A and B
27. Which statement about the main method is true?
 - A) There can be multiple main methods in a class with the same signature
 - B) JVM looks for exactly one public static void main(String[] args) to start execution
 - C) main method can be private and still run from JVM
 - D) main must always throw Exception
28. Where are static fields stored in memory in Java?
 - A) Stack
 - B) Heap (object area)
 - C) Method Area (or Metaspace in Java 8+)
 - D) Local variable area
29. What are the default values of an instance boolean and an instance int?
 - A) boolean = true, int = 0
 - B) boolean = false, int = 0
 - C) boolean = false, int = null
 - D) boolean = null, int = 0
30. What is the default value of elements in an int array ?
 - A) null
 - B) 0
 - C) false
 - D) Undefined
31. What happens if you omit the break statement in a switch case?
 - A) Compile-time error
 - B) Program terminates immediately
 - C) Fall-through occurs and execution continues to the next case
 - D) Only the first case executes
32. Is the default case in a switch statement mandatory?
 - A) Yes, otherwise compile-time error
 - B) No, it is optional
 - C) Only if using String in switch
 - D) Only if using int in switch
33. What happens if a switch executes a case and there is no matching case and no default?
 - A) Compile-time error
 - B) Runtime exception
 - C) Switch block is skipped, program continues after switch
 - D) Switch executes first case by default
34. What does the break statement do inside a loop?
 - A) Skips the current iteration and continues the next
 - B) Exits the current loop immediately
 - C) Exits the program
 - D) Jumps to the first iteration of the loop
35. What does the continue statement do inside a loop?
 - A) Terminates the loop completely
 - B) Skips the remaining statements in the current iteration and moves to the next iteration
 - C) Exits the program
 - D) Breaks the current method
36. Which class is the superclass of all classes in Java?
 - A) Object
 - B) Class
 - C) Super
 - D) String